



ISI MSQE PEA 2026 PYQ Practice Booklet

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STATSTRIVE

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Question 1

Suppose a and b are positive integers that are relatively prime, and

$$\frac{a}{b} = \sum_{n=1}^{2026} \frac{1}{n^2 + 15n + 56}.$$

Then the value of $a + b$ is

- (A) 9149
 - (B) 9229
 - (C) 8189
 - (D) 11
-

Question 2

If $x^2 - 3x + 1 = 0$, then the value of $\left(x^4 + \frac{1}{x^4}\right)$ is

- (A) 43
 - (B) 45
 - (C) 47
 - (D) 49
-

Question 3

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $f(x^3 + 1) = x^6 + 4x^3 + 8$ for all $x \in \mathbb{R}$. Then $\int_1^2 f(x) dx$ equals

- (A) $\frac{29}{3}$
 - (B) 10
 - (C) 11
 - (D) $\frac{31}{3}$
-

Question 4

Determine the value of

$$\int_0^2 |x^2 - 7x + 10| dx.$$

- (A) 8
 - (B) 9
 - (C) $\frac{26}{3}$
 - (D) $-\frac{23}{3}$
-

Question 5

Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous with $\int_0^1 f(t) dt = 1$, and let

$$P(x) = \sum_{k=1}^n a_k x^k, \quad \sum_{k=1}^n a_k = 1.$$

Then there exists $c \in (0, 1)$ such that

- (A) $f(c) = 2 \sum_{k=1}^n a_k$
 - (B) $f(c) = \int_0^1 P(t) dt$
 - (C) $f(c) = P'(c)$
 - (D) $f(c) = P(c)$
-

Question 6

Let $f : [0, \pi/4] \rightarrow \mathbb{R}$ be continuous. Then there exists $c \in (0, \pi/4)$ such that

- (A) $f(c) = \int_0^{\pi/4} f(t) dt$
 - (B) $f(c) = 2 \cos(2c) \int_0^{\pi/4} f(t) dt$
 - (C) $f(c) = \sin(2c)$
 - (D) $f(c) = \sin(2c) \int_0^{\pi/4} f(t) dt$
-

Question 7

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is continuous and

$$\int_a^x f(t) dt = \int_x^b f(t) dt \quad \text{for all } x \in [a, b].$$

Then $f(x)$ must be

- (A) A non-zero constant
 - (B) A linear function
 - (C) Identically zero
 - (D) An odd function about $\frac{a+b}{2}$
-

Question 8

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous odd function, vanishing exactly at one point and satisfying $f(1) = \frac{1}{2}$. If

$$\lim_{x \rightarrow 1} \frac{F(x)}{G(x)} = \frac{1}{14},$$

where

$$F(x) = \int_{-1}^x f(t) dt, \quad G(x) = \int_{-1}^x t |f(f(t))| dt,$$

then the value of $f(\frac{1}{2})$ is

- (A) $\frac{1}{7}$
 - (B) 7
 - (C) -7
 - (D) $\frac{7}{2}$
-

Question 9

If the correlation between X and Y is zero, then which of the following is always true?

- (A) They are independent
 - (B) There is no linear relationship
 - (C) They cannot be functionally related
 - (D) All of the above
-

Question 10

In simple linear regression $Y = a + bX$, the least squares line based on n data points always passes through

- (A) $(0, 0)$
 - (B) $(\frac{1}{n}, \frac{1}{n})$
 - (C) $(1, 1)$
 - (D) None of the previous
-

Question 11

Given sample size n , reducing the significance level α

- (A) Increases Type I error
 - (B) Decreases Type II error
 - (C) Increases power
 - (D) Decreases power
-

Question 12

If $X \sim \text{Bernoulli}(p)$, then $E(X^2)$ equals

- (A) p^2
 - (B) p
 - (C) $p(1 - p)$
 - (D) $p + p^2$
-

Question 13

For $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$ (i.i.d.), which is unbiased for λ^2 ?

- (A) $\bar{X}^2$
 - (B) $\bar{X}$
 - (C) $\bar{X}^2 - \frac{\bar{X}}{n}$
 - (D) $\bar{X}^2 - \bar{X}$
-

Question 14

A dataset consists of 9 observations arranged in increasing order $x_1 \leq x_2 \leq \dots \leq x_9$ with mean 20 and median 18. If x_9 is replaced by a very large number, which must be true?

- (A) Both the mean and the median increase
 - (B) The mean increases but the median remains unchanged
 - (C) The median increases but the mean remains unchanged
 - (D) Both the mean and the median remain unchanged
-

Question 15

A restaurant has 60% vegetarian customers (of whom 30% order dessert) and 40% non-vegetarian (of whom 50% order dessert). Given that a randomly selected customer ordered dessert, what is the probability they ordered vegetarian?

- (A) 0.375
 - (B) 0.474
 - (C) 0.600
 - (D) 0.750
-

Question 16

Suppose preferences are represented by $u(x, y) = x - \ln y$ for $x, y > 0$. Then the underlying preference relation must be

- (A) Incomplete
 - (B) Complete but intransitive
 - (C) Complete and transitive but discontinuous
 - (D) Complete, transitive and continuous but not convex
-

Question 17

Suppose there are two agents A and B with utilities

$$u_A = x_A + y_A - \frac{m}{2}(x_A - y_A)^2, \quad u_B = x_B + y_B,$$

where x_i and y_i are the amounts of goods X and Y consumed by agent $i \in \{A, B\}$. The total endowments are x and y . Initially A owns the entire endowment of Y and B owns the entire endowment of X . Then A will voluntarily transfer a positive amount of Y to B if

- (A) $m > 0$
 - (B) $m < 0$
 - (C) $y > 2$
 - (D) $my > \frac{1}{2}$
-

Question 18

Two agents A and B have utilities

$$u_A = x_A + y, \quad u_B = 2x_B + y,$$

where x_A, x_B are amounts of a private good and y is the amount of a public good. Production satisfies $y + m(x_A + x_B) = 1$, with $m \in (0, 1)$. Suppose a social planner decides that one unit should be produced. For what value of m will the resulting allocation be Pareto optimal?

- (A) $\frac{4}{5}$
 - (B) $\frac{4}{7}$
 - (C) $\frac{4}{9}$
 - (D) $\frac{4}{11}$
-

Question 19

Three consumers with valuations v_A, v_B, v_C each wish to buy one unit. The monopolist has zero marginal cost and knows the valuations but must charge a uniform price. It is given that

$$\frac{7v_B}{6} > v_A > v_B > \frac{8v_C}{5} > 0.$$

The profit-maximising price is

- (A) v_A
 - (B) v_B
 - (C) v_C
 - (D) Something in (v_C, v_B)
-

Question 20

n agents simultaneously choose private consumption x_i and public-good contribution y_i to maximise

$$u_i = x_i \left(\sum_{j=1}^n y_j \right),$$

subject to $x_i + y_i = b$, $x_i, y_i \geq 0$. As n increases, the total Nash equilibrium expenditure on the public good

- (A) Remains invariant
 - (B) Monotonically increases and approaches b
 - (C) Approaches $\frac{b}{n}$
 - (D) Monotonically decreases and approaches 0
-

Question 21

A and B are neighbours; the snow plough cannot clear in front of A 's house without clearing in front of B 's. Inverse demands are $P_A = 12 - q$, $P_B = 8 - q$. Marginal cost is 16. What is the efficient level of provision?

- (A) 6
 - (B) 4
 - (C) 2
 - (D) 1
-

Question 22

A lump-sum tax of Rs. 1 is paid by the buyer per unit of a competitive good. The price paid by buyers rises by 80 paise. Then

- (A) Demand must be perfectly elastic
 - (B) Demand must be perfectly inelastic
 - (C) Demand and supply elasticities must both be positive but finite
 - (D) Supply must be perfectly elastic
-

Question 23

A consumer always spends one-fourth of income on X , and the income elasticity of demand for X is 5. Is Y inferior?

- (A) Yes
 - (B) No
 - (C) Y is inferior iff $p_Y \geq 2p_X$
 - (D) Y is inferior iff $p_Y \leq p_X/2$
-

Question 24

Half the total quantity is purchased by 75 consumers each with price elasticity 2, and the other half by 25 consumers each with price elasticity 3. What is the aggregate price elasticity for the 100 consumers?

- (A) Cannot be determined
 - (B) 2.25
 - (C) 2.5
 - (D) 2.75
-

Question 25

Two countries H and F produce single goods with outputs Y_H, Y_F . Take H 's good as numeraire and p as the relative price of F 's good. Consumers in country i spend a fraction $1/4$ of their expenditure on the foreign good. E_i denotes total expenditure of country i measured in its own good. Given $Y_H = 120$, $Y_F = 100$, $E_H = 80$ and market clearing with $Y_H + pY_F = E_H + pE_F$, find p .

- (A) 8
 - (B) 6
 - (C) 4
 - (D) 2
-

Question 26

An economy has one good produced one-for-one from labour. The representative consumer has $U(C, \frac{M}{P}) = \frac{3}{4} \ln C + \frac{1}{4} \ln \frac{M}{P}$, labour endowment 50, money endowment $\bar{M} = 100$, with constant money supply $M = \bar{M}$. If the price is flexible (so labour is fully employed), the equilibrium price is

- (A) 2
 - (B) 4
 - (C) 6
 - (D) 8
-

Question 27

With the same setup as Question 26 but with price fixed at $P = 10$, the equilibrium output is

- (A) 15
 - (B) 20
 - (C) 25
 - (D) 30
-

Question 28

Solow economy with $Y = K^{1/2}L^{1/2}$, no depreciation, $n = 0.02$, savings rate $s > 0$. If the steady-state capital-labour ratio is $k^* = 9$ and the current ratio is $k = 1$, then the growth rate of $y = Y/L$ at the current date is

- (A) 0.02
 - (B) 0.04
 - (C) 0.03
 - (D) Indeterminable
-

Question 29

Two Solow economies A, B share the same technology and $n = 0.02$. They have $k_A = 3, k_A^* = 9$ and $k_B = 4, k_B^* = 16$. Then per capita output is growing

- (A) Faster in A than in B
 - (B) Faster in B than in A
 - (C) At the same rate in A and B
 - (D) At an indeterminable relative rate
-

Question 30

A two-period consumer maximises

$$U(C_1, C_2) = \log C_1 + \frac{1}{1+\rho} \log C_2, \quad \rho > 0,$$

subject to $C_1 + S = w$ and $C_2 = (1+r)S$, with $w, r > 0$. If w rises by 1% and r falls by 1%, then S

- (A) Increases by more than 1%
 - (B) Increases by 1%
 - (C) Decreases by 1%
 - (D) Decreases by more than 1%
-

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